

Accessibility Feature Implementation Within Free VR Experiences

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ABSTRACT

Virtual reality (VR) enables many exciting, immersive experiences; however, those experiences are sometimes inaccessible. Researchers have explored methods to improve VR accessibility, but it is also important to ensure that we have a clear understanding of the prevalence of current accessibility feature implementation to identify what support is less common for different impairments (e.g., vision, motor, hearing). In our small-scale study, we focused on free VR experiences since everybody can install those apps and games without any cost. We inspected 106 free VR experiences available for the Meta Quest 2 so that we could determine if relevant accessibility features were available for the modes of interaction present, as well as the overall prevalence of accessibility features across the sample of VR experiences. We found that 36.8% of our sample did not include any accessibility features, and many other VR experiences could have benefited from additional accessibility features. We make recommendations that can help to ensure future VR experiences are implemented with more accessibility in mind.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility**.

KEYWORDS

Accessible design, free software, virtual reality.

ACM Reference Format:

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1 INTRODUCTION

The HTC Corporation received a lawsuit in 2020 from a deaf man called Dylan Panarra due to no captions for its virtual reality (VR) experiences¹, thus making those VR experiences inaccessible [11]. Dylan Panarra won the case, but it raises the question: *how many important accessibility features are missing from VR experiences?*

VR has grown steadily in recent years, with over 171 million users of VR worldwide [24], and VR gaming revenue is projected to

be worth \$2.4 billion in 2024 [24]. VR's popularity means there is a rising need to make VR experiences accessible to disabled users by designing experiences to meet their access needs [40, 46, 55, 57]. While VR is a newer system within the consumer market compared to other technologies, VR creators could draw from the long history of accessible design and make use of new guidelines [34, 35, 51] to inform the design and development of accessible VR experiences.

We inspected 106 free VR experiences available for the Meta Quest 2 to determine the prevalence of accessibility features (e.g., captions, magnification, text-to-speech, etc.) that would be useful for each VR experience's design and mode of interaction. We focused on analyzing only the free VR applications available on the Quest store because the free VR applications are readily available to all individuals who own a Quest 2 headset without any additional cost of purchasing VR applications or VR games. It is important to understand the level of accessibility feature prevalence among non-profit/free VR experiences so that we can determine if designers and developers of those types of VR experiences require support. Furthermore, Quest devices are immensely popular, with tens of millions of units sold and future devices already planned [18].

We found striking gaps in accessibility feature implementation. For example, out of the 106 free titles, there were only 67 experiences that had *at least one* accessibility feature. The remaining 39 had *no* accessibility features. We provide a detailed breakdown of accessibility feature implementation in Section 4. We make recommendations on how the HCI and VR communities can work toward improving accessibility feature implementation in VR experiences. We also outline future research such as exploring the creation process of VR designers and developers to understand how they choose to implement accessibility features.

2 RELATED WORK

Our literature review focuses on highlighting work that spans key disability categories according to W3C's XAUR Guidelines, and we explore how prior work has implemented accessibility in VR experiences. See Dudley et al. [9] for a more extensive literature review on inclusivity within the broader scope of immersive technologies.

2.1 Accessibility in VR Apps and Video Game

There are specific VR guidelines as part of W3C's XR Accessibility User Requirements [51], as well as Meta Quest's guidelines for accessibility [34, 35]. Heilemann et al. [19] have worked to consolidate guidelines across both published work and vetted websites. Moreover, researchers and organizations [12] are dedicated to advocating inclusivity and improving accessibility in this domain. According to W3C's XAUR Guidelines [51], there are 6 main categorizations for disability. These are visual, audio, mobility, speech, neurological,

¹We will refer to VR apps and VR games as VR experiences for simplicity.

and cognitive disabilities. In this section, we explore prior research that focuses on each of these categories (we combine two of them).

2.1.1 Visual. Research has investigated using haptic feedback, sound and visual filters to create experiences for visual disabilities. We have seen how augmented reality systems have incorporated visual accessibility [13, 43]. For instance, SeeinVR [59] aims to solve the issues of visual accessibility as an afterthought by making minor accessibility implementations to existing features. Prior work [23, 25, 58] also shows how virtual reality can utilize haptics and space to help provide information about position and location to blind and low vision (BLV) users. Motion sickness can be heavily induced by visual stimuli, with prior work discussing effective ways to handle triggers [20, 44].

2.1.2 Audio. Audio-related disabilities are generally focused on utilizing subtitles, labels, visualizers, or volume adjusters to help elevate critical audio over others to help curb the inaccessibility of experiences. We have seen video games like Fortnite [14] that use visualizers and subtitles, as well as allow players to adjust the volume in their experience to improve access for deaf and hard of hearing (DHH) users. This is an excellent implementation of accessibility as it helps solve audio-related disabilities with little to no hassle for users.

Prior research has looked at understanding how to meet the needs of DHH people [41]. SoundVizVR [28, 29] is a tool that can be used to help visualize audio in virtual reality and serves as an easy inclusion for developers while working on their experience. Jain et.al. [22] also explored making every day sounds more accessible to DHH users in VR users by using haptic and visual feedback. Additional work has looked at creating a taxonomy of sounds in VR as well as conducting an analysis of sound design in VR [21].

2.1.3 Mobility. Another area that has been researched is mobility within the space of VR [26]. Prior work showcases the manner in which VR can benefit rehabilitation or be fun experiences for wheelchair users and people with motor disabilities [7, 15, 27, 30, 31, 36, 37, 42]. WalkIn VR [8] is an application on Steam that allows people with a mobility impairment to adapt VR games (e.g., WalkIn VR provides assisted gameplay for wheelchair users to access positions and locations in the virtual space by adjusting their height and range of movement across applications).

2.1.4 Neurological and Cognitive. Similar to the easy-to-use functionality of video games, virtual reality can incorporate easy-to-follow tutorials, training, and readily available help features to decrease the cognitive load on users [2]. Apart from these features, the inclusion of assisted gameplay modes or providing simpler interactions can significantly aid in these experiences. Traditional video games tend to offer more variety in terms of accessibility features for neurological and cognitive disabilities [56]. We have seen this with video games offering easier game modes; something that is yet to be incorporated in a lot of commercial VR titles.

2.1.5 Speech. It helps to have alternate modes of communication in the form of quips or gestures. VRChat allows users to use emoji, emotes, and gestures in the form of Helping Hands [1] to communicate without the need for text-to-speech or additional physical

equipment. The issue with VRChat is that the feature is not integrated into the experience and requires users to access it externally. Prior work associated with speech impairments explores using virtual reality for speech therapy [5, 16, 33]. It is important to acknowledge that accessibility features for speech are also useful to users who may have anxiety [10, 17] or who have a stutter [52].

3 ACCESSIBILITY FEATURES ANALYSIS METHOD

Our research involved an analysis of the accessibility features present for the free VR experiences (both applications and games) on the Quest 2. Our analysis took place May to July 2023.

3.1 Apparatus and VR Experiences

We selected the Quest 2 due to its popularity as a VR headset with its cable-free setup. The titles that we investigated were available for free (including demos and trials of paid VR experiences). We retrieved data from the Quest Store and found a total of 121 free experiences². However, after inspecting them individually, only 106 of these VR experiences were actually available for download and use. The remaining 15 apps still have active pages on the Quest Store but are either paid or do not allow users to download or add them to the library.

3.2 VR Experience Inspection

We systematically reviewed free VR experiences and checked them against the accessibility feature list as mentioned in Table 1. We categorized the accessibility features using guidance from W3C's XAUR [51], Meta [34, 35] and Heilemanns et al. [19] (Appendix A provides additional details on category definitions and identifying common criteria among different guidelines).

To provide an example, 'Elixir' is a fantasy game that explores an uncharted magical portal. The user takes on the role of an apprentice who is expected to comply with the instructions given to them and perform various tasks such as making magic potions, interacting with magical creatures, and exploring the usage of the sorcerer's apparatus. The apprentice receives verbal instructions, but there are no subtitles or language configurations, which means those missing accessibility features make the game less inclusive. Furthermore, some of the interactions in the experience are based on color, but there are no controls or accessibility features for people with color blindness. There were other missing features, but the brief example above is to provide clarity on our process.

We based the decision of whether an application should have included a particular accessibility feature based on the type of design style and interaction modes the VR experience presented. For example, all text-based applications, such as VR experiences designed for education, creativity, and productivity, should have the magnification feature because they are predominantly text-focused, and users would benefit from a magnification feature to support reading text. One example is Nanome, which is a VR experience allowing users to interact with molecular data. Different elements on the screen had labels containing descriptions and information about the element, but not all of the text was legible. A magnification

²Data of Quest 2 VR titles taken from <https://vrdb.app/quest/>

feature would improve accessibility. On the other hand, some VR experiences did not involve textual elements and would not require text magnification. For example, we accessed the demo version of Beat Saber, which is a game where you slash objects to the beat of music. This game does not involve any textual interactions and is based on movement gestures. The objects that the participant interacts with are of a significant size; therefore, we determined that magnification was not currently an essential missing feature.

Our data collection methodology consisted of the following steps:

Step 1: We made a list of all the free games available on the Oculus store that were also compatible with Quest 2 by reading the description and manually downloading the qualifying games.

Step 2: After downloading the application, we launched it from the apps menu in the Quest Home.

Step 3: We followed the application or tutorials for applications that had the option, and in cases where no help was provided, we tested the VR experience until we understood.

Step 4: We then reviewed menus that could usually be accessed by glancing briefly at the hand or controller. The menu contains different features and settings available within the application.

Step 5: The main menu sometimes had accessibility features hidden under separate categories such as General, Audio, Video, Control, etc. Therefore, we carefully assessed all the menu items to identify accessibility features present for each application. We documented our findings by recording specific accessibility features, if any, and also noted if there were no accessibility features. To verify if the VR experiences included accessibility features, the first two authors performed a systematic review of the games and applications. Our decision on whether an experience would benefit from a particular accessibility feature was made based on the kind of interactions that the experience required. For example, if the VR experience required decisions based on color, we would note that the VR experience would benefit from a color blind mode; if a VR experience allowed multiplayer, it would benefit from speaker labels, etc.

4 ACCESSIBILITY FEATURES FINDINGS

4.1 App Specific Findings

We explored various VR experiences covering different genres such as social, media & streaming, games (action, arcade, casual, etc.), productivity, documentary, animation, health and fitness, education, 2D, utilities, etc. Out of the 106 free VR experiences, seven were demos, and two were free trials for subscription-based applications. Nine VR experiences were 2D, i.e., they were not 360-degree experiences and could be accessed within the Quest home screen.

We found that only 67 VR experiences had at least one accessibility feature, while 39 VR experiences offered no accessibility features. Out of the 67 experiences, we note that the majority (41) only had between 1-3 features. These were mostly tutorials, language, and help features and rarely had an emphasis on other accessibility categories. Figures 1-3 provide an example of how accessibility features were presented within the VR experiences. VR Chat was the only experience that had as many as 18 accessibility features in a single VR experience. The remaining experiences had less than 10 accessibility features, with 25 falling in the range of 4-10 accessibility features. Overall, there was a discrepancy in the number of

accessibility features implemented and the number of accessibility features required by an experience (as summarized in Table 1).

4.2 Visual Design Settings

Through our investigation, we identified a predominant issue with visual design settings, which would affect the experience of people with vision impairments. We found that only one (out of 18) experiences had the option to adjust the font size, and none of the experiences had the option to adjust font style or color. Visual filters (e.g., high contrast mode) were available in two of the VR experiences (out of 48 VR experiences that should have offered filters, as shown in Table 1), while two of the VR experiences allowed for magnification (out of 20 that should provide a magnification feature), and three (out of 25) had options for HUD adjustment. None of the VR experiences we tested had a color blind mode, which was a significant issue because 81 VR experiences had some form of interaction that relied on color perception.

Magnification is another feature that defines the accessibility of an experience, and we noted that only two VR experiences included the option (out of 20 that would benefit from magnification). We found that VR experiences designed for education, creativity, and productivity would greatly benefit from this feature as they were designed for users to perceive information, especially text, and also within virtual meeting and collaboration settings. For example, integrating a magnification feature would allow a user to view the content on a whiteboard more clearly.

4.3 Motor Settings

We noted that some experiences had haptic feedback on the controllers, which made the experience more immersive. However, 79% of the experiences that needed this feature did not have it (Table 1). Features like one-handed and left-handed modes, which aid in making experiences accessible to people with motor difficulties, were present in very few experiences. None of the VR experiences that would benefit from it had input control sensitivity (0/30), and only one experience included a remapping feature (1/39). Aim assist is a common feature among action and shooting games designed to help users aim accurately at the target. While 43 VR experiences required users to aim at a target, only one of those VR experiences had this feature. Some experiences involved intense physical movements but did not have any motion settings, which made the experience inaccessible to users with physical impairments. PokerStars VR and Rec Room were the only experiences allowing the user to select the dominant hand before starting the experience. This feature is an excellent example of inclusive design as it allows all users to choose their dominant hand for play rather than presuming a particular hand to be dominant.

4.4 Audio Settings

We noted that 20 VR experiences offered the option to adjust the volume levels of various sounds out of the 91 VR experiences that should include this accessibility feature. Furthermore, missing features included sound visualizers and descriptive subtitles. Descriptive Subtitles are the audio descriptions of the visual elements of a scene, such as a person entering a room, facial expressions, and other relevant details that define the scene. 15/20 did not provide

Table 1: Accessibility Features Categories: This table consists of a list of categories that were based on the guidelines offered by W3C’s XAUR [51], Meta [34, 35] and the consolidated list of guidelines from Heilemann et. al. [19]. The numbers within the brackets indicate the total number of VR experiences with the accessibility feature present against the number of experiences that could benefit from the corresponding feature. Note: Abbreviations in this table are as follows: Sens. = Sensitivity, Subs. = Subtitles, Alt = Alternative, M.S. = Motion Sickness. See Figure 4 for an alternative presentation of our data in bar chart form.

Vision	Motor	Audio	Communication	General
Colorblind Mode (0/82)	Input Control Sens. (0/30)	Subtitles (5/81)	Speech to Text (17/51)	Difficulty Levels (3/12)
Visual Filters (2/48)	Remapping (1/39)	Closed Captions (3/20)	Text to Speech (0/35)	Tutorials (44/80)
Font Size (1/18)	Alt. Control Inputs (3/41)	Descriptive Subs. (0/23)	Alt Text Input (4/15)	Navigation Cues (1/33)
Font Style (0/18)	Haptic Feedback (9/43)	Speaker Label (1/25)		Language (12/67)
Font Color (0/18)	One-handed Mode (2/39)	Volume Control (20/91)		M.S. Mode (10/44)
HUD Adjustment (3/25)	Left-handed Mode (8/39)	Sound Visualizers (0/47)		
Magnification (2/20)	Aim Assist (1/43)			

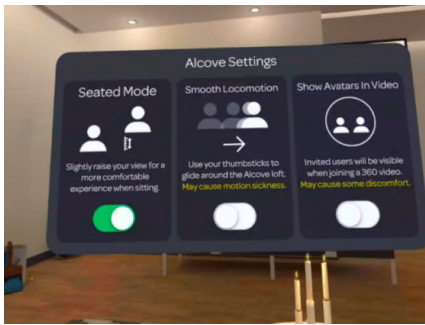


Figure 1: Alcove allows users to customize seated mode by adjusting the height and also allows smoother locomotion using a glide motion.

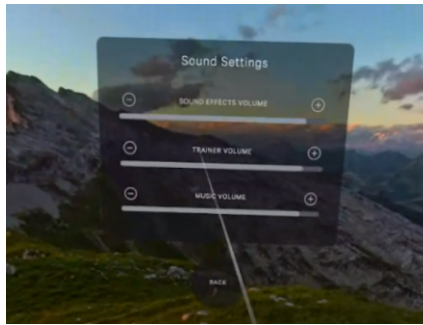


Figure 2: Audio settings in Supernatural where users can manually adjust the volume levels of different elements for enhanced immersion.

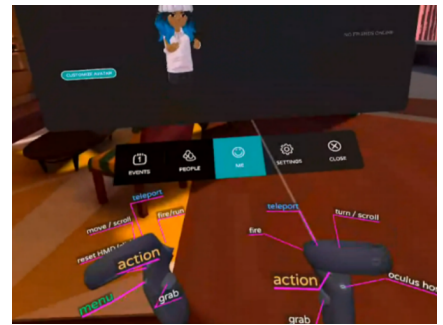


Figure 3: AltSpaceVR provides users with an onboarding experience by drawing labels connected to various controller buttons.

features to enable captioning services, subtitles, or descriptive audio. On the other hand, closed captions are the transcriptions of dialogues along with other elements such as speaker labels and background music. We noted that only 15% (3/20) of the VR experiences that would benefit from closed captions offered them. Supernatural (Figure 1) shows how users can adjust sound effects, music, and the volume of the trainer. Channels and spatial audio control can often be useful for people with hearing loss and who might want to prioritize some specific sounds over others (e.g., clear character voices and reduced background music). Since Supernatural is supposed to help users focus on health and fitness, the volume of the trainers' voice is a high priority. Conversely, users who cannot perceive haptic feedback from completing tasks in the mini-games can benefit from prioritizing sound effects.

4.5 Communication Settings

Speech to text (STT) is a default feature on the Quest 2 and allows users to enable and disable their microphones at will. Our audit found that STT was the third most prominent feature, present in 17 experiences (out of 34 that could have benefited from the feature). 14 out of the 17 experiences that offered STT were social VR applications. Experiences such as Horizon Worlds (including Horizon Venues) offer safety features and voice-based input. This

included a feature that would garble voices for users outside a particular radius, thereby giving them the confidence to communicate without worrying about people overhearing conversations across a virtual environment. We also found 35 experiences that should have included a text to speech setting, but none of those VR experiences included one. Only 27% of VR experiences that required text input had an alternative text input, and only 33% of VR experience with spoken audio offered the option of subtitles. Alternative text input refers to how a user can input text to a machine without using a controller (e.g., through voice).

4.6 General Settings

4.6.1 Tutorials and Help. The most common features, which can be a form of accessibility support, were help and tutorials/training modes, which were included in 37 and 35 VR experiences, respectively (44 in total, not including duplicate features). Out of these, 28 experiences had both tutorials and help features. The two features are similar. However, the help feature serves as a way to access information about the experience (controls, functions, etc.) and can be accessed at any point in time. The guidelines we referred to suggest the use of training modes or tutorials to help users with neurological and cognitive disabilities with interactions within the experience [19, 34, 35, 51]. We also note that the inclusion of this

feature may appear most prominently compared to other accessibility features in our VR experience sample because VR is considered to be a relatively new platform for many people, therefore prompting creators to add tutorials to make the experiences easier for newcomers. Regardless, this feature should be more widespread and incorporated into more experiences.

Although 44/106 VR experiences had tutorials, we determined that 80 VR experiences should have offered tutorials to give the user a brief introduction to the experience and the controls. For example, Arthur was a social and productivity app that offered a virtual office space for users to meet, collaborate, and work. However, the experience did not offer any tutorial that would help the user understand the features and controls of the app. The remaining VR experiences consisted of documentaries or educational experiences that did not involve complex actions or controls. For example, Notes On Blindness is a documentary created by a man called John Hull who has total blindness. The VR experience was only audio and visual with no interactions required from the user.

4.6.2 Language. Language settings are discussed as an accessibility feature in current guidelines to include audio and textual content in the language of preference of the user [19, 51]. 12 VR experiences offered language options, which account for only 18% of the VR experiences that could have used language options; however, those options varied. A few offered as little as two or three language alternatives, whereas some experiences offered a wide variety of languages.

4.6.3 Player Modes and Motion Sickness. We found 10 VR experiences with accessibility features for player modes (one-handed and left-handed modes) and 10 VR experiences with a motion sickness mode. These features were almost always coupled together in experiences for the sake of overall player mobility. Alcove (Figure 1) depicts locomotion and seated modes. Supported player modes are usually specified for each VR experience on their Quest 2 store page. This has been included to improve the accessibility for wheelchair users without additional add-ons or features. Usually, the two major modes of movement were in the form of teleportation or smooth movement (as seen as an option on Alcove). An outlier to both of these was the Under Presents, which involved a warping movement that was done primarily by pulling and tugging through the environment. Considering game mechanics, this was rather novel; however, this would likely be inaccessible to some users with motor impairments. In addition to this, comfort modes were a feature to address motion sickness (tunneling, vignette effects) by making movement and motion in VR easier on users. This, however, was implemented only in 23% of the experiences. Finally, general settings, such as adjusting the difficulty level and navigational cues, were often absent.

5 DISCUSSION

Our work provides insights into current accessibility feature implementation among free VR experiences for the Meta Quest 2. Through our analysis, we found minimal emphasis is placed on incorporating accessibility features.

One reason for this might be that despite guidelines existing, the accessibility guidelines provided by the three sources (W3C's

XAUR [51], Heilemann et al. [19], and Quest 2 Guidelines [34, 35]) are for consideration without much consequence for not following them. Furthermore, there could be challenges related to not understanding how best to implement suggestions. Finally, some creators may not know there are guidelines out there. Inaccessibility in older and more traditional platforms such as web [4, 6], mobile [3] and video games [54] has been commonplace, and it seems the trend continues in this space of immersive technology.

We draw upon prior work that critiques accessibility guidelines in related platforms [32, 38, 39], which often results in designers looking for improved guidelines or alternatives [45, 47, 48], and stress that current VR guidelines may need to go through a more critical review for improvement. Future guidelines must be well defined to be tightly enforced, leading to the proper incorporation of accessibility features.

Considering that we are in the nascent stages of VR accessibility, it is important to have clear instructions and support for including accessibility. Existing VR guidelines discuss important topics associated with accessibility but do not tackle the implementation aspects of it. Having targeted accessibility guidance will aid VR designers and developers in creating more accessible experiences rather than incorporating accessibility as a post-hoc fix [28, 49, 50, 53].

5.1 Recommendations

5.1.1 Establishing Specific Guidelines. We recommend that accessibility and HCI researchers focus on refining accessibility guidelines for VR experiences. Existing guidelines, such as the Meta Quest [34, 35], XAUR [51], and the consolidated guidelines from Heilemann et al. [19], might be too vague. We noted missing accessibility features in the free VR experiences despite the availability of guidelines, thus suggesting a disconnect between the resources available and the output of creators. It is also not sufficient to suggest that creators consider adding features but to also provide directions with examples on how these features can be implemented.

5.1.2 Reforming VR Work Culture. There are opportunities for HCI researchers to investigate methods for improving the working culture that currently affects the extent to which accessible VR experiences are created, particularly for people working on free VR experiences. Many free VR experiences can come from small companies or even independent creators who are likely to have fewer resources compared to large companies. Prior work has highlighted access to resources has an effect on how likely accessible design is to be implemented [4]. Therefore, it is important for HCI researchers, especially those working in the domain of Computer-Supported Cooperative Work (CSCW), to explore and understand the workplace settings of VR creators to identify issues contributing to what we found.

Guidelines and awareness of accessibility go hand in hand. Although including less vague guidelines can truly help mitigate the issue we have at present, it is important to advocate for accessibility as well as involve creators with disabilities in the process of developing and designing VR experiences. It becomes extremely important to start at every level by providing awareness and education to designers and developers. Future work can focus on

building awareness that starts with exposing creators to the benefits of bringing inclusivity into the forefront of VR and, therefore, promoting the implementation of accessibility features.

5.1.3 Accessibility Friendly Tools for VR. We also recommend that accessibility and HCI researchers continue to explore creating accessibility tools to support VR creators. The focus should be on enhancing the existing tools to support the creation of accessibility features for VR experiences.

We could also achieve greater inclusivity by having the leading organizations in the creative VR space include accessibility into pre-existing software tools and systems, which can then be made available to other VR creators who may be working independently or at smaller organizations.

6 LIMITATIONS AND FUTURE WORK

We identified the prevalence of accessibility features within VR experiences. A recent lawsuit drew attention to the lack of captions offered by HTC [11], but there was no systematic review showcasing the prevalence of other important accessibility features. We found that very few accessibility features are included within VR experiences; however, we did not assess whether those included features were implemented effectively. It would be beneficial in future work to audit whether those accessibility features actually translate to an accessible experience for disabled people.

Furthermore, we utilized the Meta Quest 2 for our work. Our justification was that it is a modern, affordable, and popular consumer product. However, it would be beneficial to extend our analysis to VR experiences on different VR platforms (e.g., Rift, Valve Index, HTC Vive, etc.).

We were unable to test seven demos and free trials that were a part of our study as they required a purchase to access the full version. However, we noticed that they did not offer too many accessibility features. This does lead us to question whether accessibility features are put behind a paywall—although we expect this to be unlikely based on our knowledge of accessible design in other digital domains. Interviews with VR creators, including those from large studios, may be useful. In the future, we plan to expand the scope of our study to understand the process that VR creators follow while designing and developing applications. This will help us understand how and when they look into incorporating accessibility features into their workflow and how they prioritize the accessibility features.

7 CONCLUSION

Virtual reality (VR) is a popular technology that continues to establish itself within society. However, to be truly successful, it needs to be accessible to all users. Prior work demonstrated current issues surrounding the accessibility of VR and suggested solutions to specific technical issues (e.g., providing visual cues for sound), and we build upon this work by investigating the prevalence of accessibility features within current free VR experiences. We contribute a manual review of accessibility features within 106 VR experiences and demonstrate striking gaps in accessibility feature implementation. We discuss factors in the design and development process that may contribute to these issues, and we make recommendations on how to address these issues through future work.

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A GUIDELINE DETAILS

A.1 Visual

Color blind Mode: Certain or all interactable objects change in color to help with specific color vision impairments (e.g., Protanopia, Deuteranopia, Tritanopia). In rare cases, this would apply as an overhaul to the entire application/game.

Visual Filters: To help locate individual objects in the scene and is similar to High Contrast Mode where the colors are in heavy contrast with one another for objects and items to be easily visible.

Text Adjustment: This refers to either subtitle text or text in the app/game, which can be adjusted for improved legibility. This could be further categorized into 1) Font Size, 2) Font Color, 3) Font Style.

HUD Adjustment: The Heads Up Display can be adjusted either in terms of color, size or reorganization of the content on it.

Magnification: To magnify objects that are not easily visible from the user's perspective. For VR, this may or may not be applicable as a lot of VR apps allow you to physically move closer to the object making it easier to view the text on it. That being said, magnification options are always helpful to read content.

A.2 Motor

Input Control Sensitivity: To alter the sensitivity of the controllers or whatever form of input the user is using.

Remapping: Changing the keybinds to make it customized for the user, especially useful for those with motor impairments.

Alternative Control Methods: Having the option to use more than one control method (apart from default controllers) One Handed Mode Allowing users to just use a single hand for any type of control/input.

Haptics Adjustment: Any options associated with adjusting the intensity of the haptic feedback or altering the type of haptics the user experiences during the course of the game.

Aim Assist: Specifically for Combat-based games that require precision in targeting virtual objects in the scene. Aim Assist focuses on helping users complete combat tasks without removing the fun aspect for those who have accessibility needs.

A.3 Audio

Subtitles: Providing captions for sounds being played. This can be further categorized into Cinematic/Descriptive Captions and Speech Captions. The former describes sounds and noises that are non-verbal in nature.

Speaker Label: Specific Labels for each individual speaker, either color-coded or with the name to provide some distinction.

Volume Control: Adjustments for volume specific to certain sounds, such as background music, sound effects, speech, etc.

Visualizers for Sounds: Sound visualizations localized to the source and presenting some form of alert on the screen for users to be informed of in-app sounds.

A.4 Communication

(for Multiplayer or Voice Recognition based Activities)

Speech to Text: Converts incoming speech to text and presents it on screen.

Text to Speech: Converts text typed out by the user into speech for ease of communication with others.

Alternative Text Input: Having other ways of communicating information - emoji, quips, etc.

A.5 General

Difficulty Levels: Having Multiple levels of difficulty ensures users can appreciate the game experience if they struggle with performing at a higher level of precision/accuracy due to certain disabilities.

Tutorial and Training Mode: Training grounds for users to test out the features and can learn the controls at their own pace.

Language: Translation options for text being shown in the scene (subs or otherwise). Can be an overhaul or specific to just subtitles alone.

Motion Sickness Mode: Eliminates potential triggers for Motion sickness by curbing the motion blur effect.

Navigation and Traversal Cues: Providing Audio/Visual hints or markers to help with navigation through the app/game.

Table 2: An example of a categorization for our accessibility features includes a reference to a section of the guidelines of W3Cs XAUR [51], Meta VRC [34, 35] and Heilemann et. al. [19] (Note: D.C. = Disability Category, XAUR = W3C XAUR Guidelines, Meta VRC = Meta’s VRC Accessibility Guideline Checklist, Meta Des = Meta’s Accessibility Design Document, HM = Heilemann et. al. Comprehensive Guidelines)

D.C.	Title	Description	XAUR	Meta VRC	Meta Des	HM
Visual	Color Blind Mode	Objects change in color to help with color vision deficiencies	4.6 Color Changes	VRC Quest Accessibility 6	Accessible UI Design	Look and Design

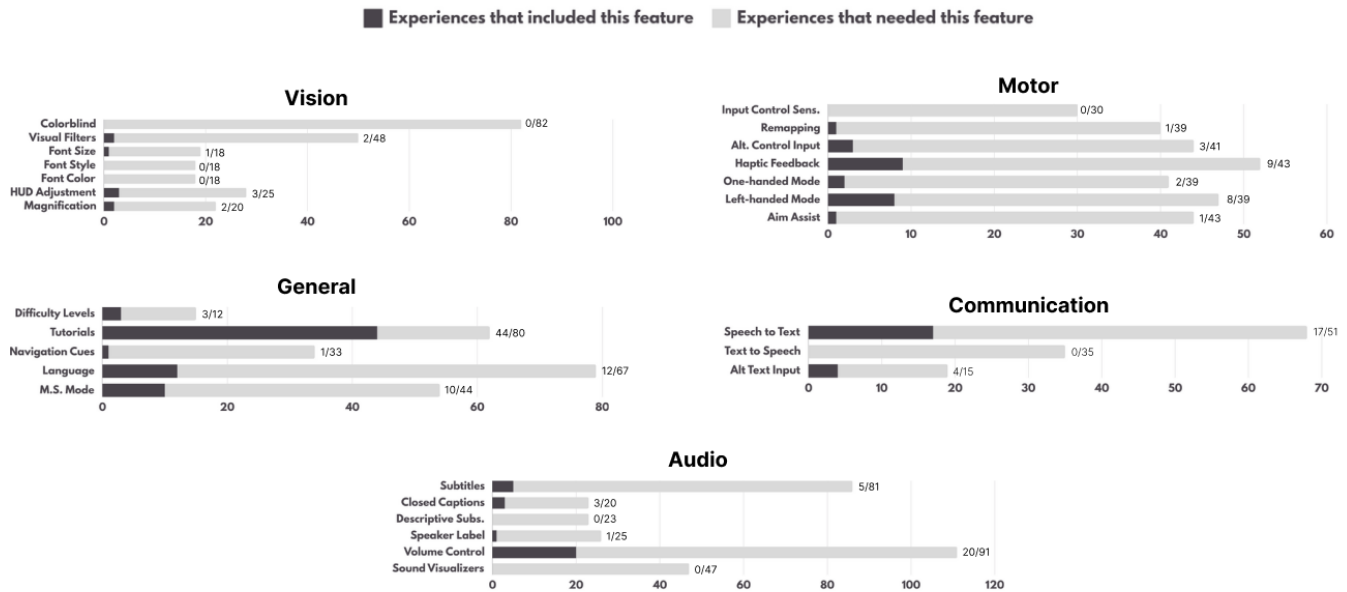


Figure 4: Bar charts for each accessibility feature category.